

Multi-objective Route Optimisation for Electric Vehicles: A Weighted A* Approach

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Abstract—Electric Vehicles (EVs) are considered the best transition from conventional fuel-based vehicles to battery-powered vehicles, as they are not only environmentally beneficial but also more economical than fuel-based vehicles. However, various limitations, such as limited driving range, long charging times, increased battery degradation, energy losses during braking, and inefficient route planning, hinder widespread adoption. As a result, route planning for EVs becomes significant. Currently, routing approaches mainly focus on distance and travel time, without considering energy utilisation in EVs. Hence, a multi-objective (MO) approach is required that considers energy consumption, travel time, cost and battery state of charge (SOC). In this paper, we have focused on route optimisation for EVs by integrating various factors: terrain, traffic conditions, and regenerative braking (RB). We implemented a heuristic method called the weighted A* algorithm, which provides faster solutions and handles the changing conditions effectively. The comparison results with other algorithms show that weighted A* achieves up to 32.3% MO cost reduction, making it suitable for EV navigation and the Intelligent Transport System.

Index Terms—Electric Vehicles, Regenerative braking, State of Charge, Graph Optimization, Heuristic Search, Path Planning, Energy-Optimal Path

I. INTRODUCTION

According to NASA, Earth's temperature had increased by approximately 1.1° Celsius above pre-industrial levels by the early 21st century. By the mid-2020s, this rise had intensified to about 1.4°, highlighting the increase in global warming and climate change [1]. As the transport sector is one of the largest contributors to greenhouse gas emissions worldwide, reducing vehicle emissions is a top priority. This situation underscores the urgent need for sustainable transport solutions and accelerates the shift from conventional to electric vehicles (EVs). However, the integration of EVs introduces challenges related to traffic conditions and the coordination of charging infrastructure; therefore, Intelligent Transport Systems (ITS) and energy management have become key components of e-mobility to address these challenges. Today, EVs are used everywhere, from public transport and battery-powered bikes [2], to electric robots [3]. EVs rely on electrical energy, which is stored in batteries, and their efficiency is strongly influenced by traffic density, speed variation, RB, and environmental

conditions [4]. Even with the higher energy efficiency and low carbon emissions, the EV market remains inferior compared to conventional vehicles. According to recent global EV sales data, about 1 in 5 (22%) of new cars sold worldwide in 2024 were EVs [11]. This shows that many people are hesitant to use EVs due to limited battery capacity and charging time. When the battery State of Charge (SOC) is low, and the charging point is located at a distance, an optimal path is required that minimises the energy consumption.

Degradation of the battery over time imposes constraints on route feasibility, leading to inaccurate range estimation and route failures in the real world [5]. These challenges indicate that EV energy consumption is influenced not only by vehicle technology but also by overall e-mobility and route optimisation strategies, making energy efficiency an important factor with economic and operational implications. Consequently, reducing energy consumption leads to less frequent charging and longer battery life, thereby lowering total cost [6]. Despite the increasing popularity due to their environmental benefits and technological advancement, EVs suffer from limited driving range [7] and battery constraints, which hinder their reliability and convenience [8]. Insufficient number of charging stations and their uneven distribution are also major concerns [9], along with uncertainties related to charging time, location, power demand, charging expenses, travel time and environmental factors [10].

In addition to efficient routing strategies, the concept of RB is crucial for the energy recovery mechanism in EVs, which helps in improving the overall energy efficiency. When the vehicle moves downhill or decelerates, the electric vehicle motor transitions into generator mode, where the motor generates back electromotive force that opposes the rotor motion and converts kinetic energy into electrical energy. When the generated voltage exceeds the battery voltage, current flows back into the battery. This process produces braking torque while simultaneously recovering electrical energy. In this way, RB helps in effective energy recovery [12], [13]. The intensity of braking depends upon the speed, slope of the road, and urban driving conditions. When the battery approaches full SOC or when the braking requirement exceeds the regenerative

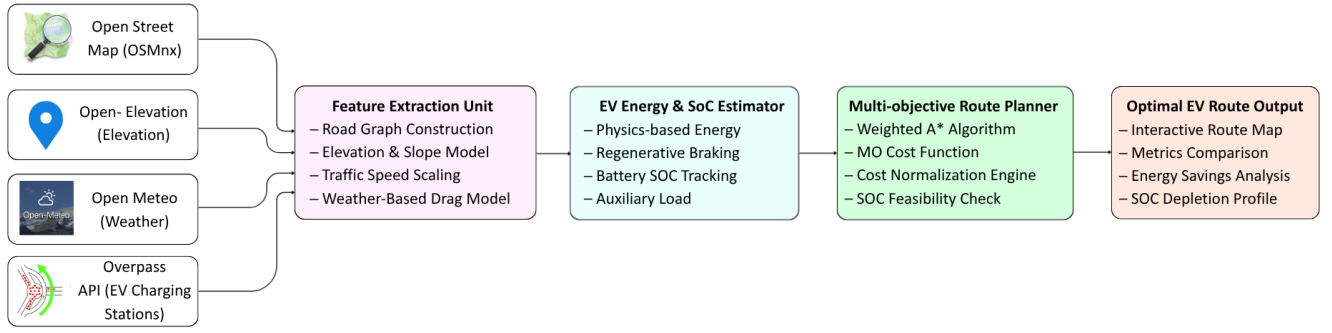


Fig. 1. System architecture of the proposed framework

capability of the electric motor, its effectiveness is reduced. Despite this limitation, RB can recover a significant portion of energy that otherwise would have been converted into heat [14] and helps in extending the driver's range.

Since the effectiveness of RB is highly dependent on road and driving conditions, route optimisation becomes a critical component in maximising the benefits. Uphill increases the energy demand, while the downhill segment provides opportunities for energy recovery through RB [15]. Traditional route planning algorithms primarily focus on reducing travel time or optimizing distance without considering energy efficiency. Route optimisation problems are addressed using methods such as the Dijkstra algorithm [16] or the Bellman-Ford Algorithm [17], which aim to determine the shortest path by considering distance or travel time. However, these algorithms overlook factors such as traffic congestion or user-specific constraints, leading to suboptimal routing decisions in a practical transportation system [18]. While these methods are useful for basic routing, they face difficulty in MO problem formulation.

In such cases, heuristic algorithms are used to overcome the issue, such as A* Algorithm, which adds directionality in the searching process of Dijkstra's algorithm and accelerates the searching [19], ant colony optimisation, which is inspired by the behaviour of ants to identify the efficient path [20], and the genetic algorithms, where candidate's route is gradually improved over successive generations [21]. These methods enhance flexibility by finding optimal solutions in complex environments. Accordingly, a charging station selection scheme using the A* algorithm and Simple Additive Weighting (SAW) was developed to improve routing under unexpected road conditions [22], [23]. Several other techniques have also been used for the route optimisation, including Particle Swarm Optimisation (PSO), MO evolutionary algorithm, model predictive control (MPC), and a reinforcement learning-based approach for handling the dynamic traffic and charging constraints [24].

In a practical EV routing scenario, route selection must balance travel time, energy usage, dynamic traffic conditions, charging cost, battery SOC and weather conditions. Reducing travel time helps in increasing the energy usage due to higher speed or congestion, whereas optimising energy efficiency can result in the selection of longer routes or the need for inter-

mediate charging stops [25]. Various charging strategies and battery degradation help in considering the further influence of route feasibility and operational cost [26]. EV routing is an MO optimisation problem that considers travel time, energy consumption, charging cost, and path distance. MO optimisation generates a solution like Pareto-optimal, which helps in capturing the trade-off among conflicting criteria and helps in decision-making under dynamic traffic and environmental conditions [27].

In this paper, we have implemented weighted A* algorithm for MO routing, enabling a balance between optimality and computational efficiency, considering factors such as traffic conditions, road quality, terrain, weather and RB. This dynamic model helps in efficient routing, reducing energy consumption, travel time and overall cost.

II. PROBLEM FORMULATION AND METHODOLOGY

The route optimisation problem aims to find the best route for EVs that considers multiple objectives, including (i) minimising energy consumption, (ii) reducing travel time, (iii) lowering financial costs and (iv) finding the shortest distance. First, we model the system and present four different objective functions, then we formulate an MO EV routing function. Finally, we present the solution to this optimisation problem using the weighted A* Algorithm.

TABLE I
VEHICLE AND SIMULATION PARAMETERS

Parameter	Symbol	Value
Vehicle mass	M	1800 kg
Air density	ρ_a	1.225 kg/m ³
Drag coefficient	C_d	0.29
Frontal area	A_f	2.2 m ²
Rolling resistance	f_r	0.015
Wheel radius	r_d	0.30 m
Wind speed range	V_w	0–10 m/s
Mass factor	δ	1.05
Drivetrain efficiency ^a	η_{drive}	0.90
Auxiliary load	P_{aux}	600 W
Battery capacity	E_{bat}	60 kWh
Initial SOC	SOC_0	80%
Minimum SOC	SOC_{min}	15%
Electricity price	P_{elec}	\$0.20/kWh
Regen efficiency	η_{regen}	0.60
Max route distance	S_{max}	300 km
Max travel time	T_{max}	3 h

EV parameters and their values used for this study are summarised in Table I. These parameters include vehicle mass, rotational effects, aerodynamic losses, and battery limitations, which serve as fixed inputs to the modelling described further.

A. Objective Functions Definition

In this study, the road network is modelled as a directed weighted graph $G = (V, E)$, where vertex V denotes the set of road intersections and E represents the set of directed edges corresponding to road segments. Each edge $e \in E$ connecting nodes u and v is defined as:

$$e = (u, v, l_e, s_e, g_e, \theta_e, \tau_e) \quad (1)$$

where l_e denotes the road segment length (in meters), θ_e represents the edge heading angle, g_e denotes the road gradient (slope), s_e represents the nominal vehicle speed (in m/s), and τ_e denotes the traffic speed adjustment factor.

We now derive four different objective functions for the energy consumption, travel time, cost, and distance. First, we calculate the force equation and derive the energy consumption for the edge e of the road network.

Newton's second law of motion can express the longitudinal dynamics of an EV travelling on a road segment (l_e) with an inclination angle θ as:

$$\delta M \frac{dV}{dt} = F_t - (F_{\text{roll}} + F_{\text{drag}} + F_{\text{grade}}) \quad (2)$$

where M is the mass of the vehicle (kg), V is the speed of the vehicle (m/s), and δ is the rotational mass correction factor due to the inertia of wheels. Rolling resistance friction caused by tyre-road interaction can be expressed as:

$$F_{\text{roll}} = MgC_r \cos \theta \quad (3)$$

where C_r is the rolling resistance coefficient, usually ranging from 0.010 to 0.020 for modern radial tyres on paved roads. The component of gravitational force acting along the direction of travel due to road inclination theta is given as:

$$F_{\text{grade}} = Mg \sin \theta \quad (4)$$

For uphill segments, this term represents the extra energy required to gain the elevation, while the downhill segments enable energy recovery using RB. Aerodynamic drag with weather effects is calculated by considering relative wind velocity, which is given by:

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A_f v_{\text{rel}}^2 \quad (5)$$

where the relative wind velocity is:

$$v_{\text{rel}} = v_e + v_{\text{wind}} \cos(\phi) \quad (6)$$

Here, v_e is the vehicle speed, v_{wind} is the magnitude of wind speed, and ϕ is the angular difference between wind direction and road heading. This formula considers both headwinds

($\phi \approx 0^\circ$, increasing drag) and tailwinds ($\phi \approx 180^\circ$, decreasing drag). Air density ρ varies with temperature according to:

$$\rho = \rho_0 \frac{T_0}{T} \quad (7)$$

where $\rho_0 = 1.225 \text{ kg/m}^3$ at reference temperature $T_0 = 298.15 \text{ K}$. Temperature changes can alter air density by $\pm 10\%$, directly affecting energy consumption.

The traction force, F_t , is the net longitudinal force that the motor applies to the wheels. This force can be positive or negative depending on the vehicle's operating mode. During RB, the motor operates in generator mode and creates a negative traction force, where η_{regen} represents the regenerative efficiency, indicating the fraction of braking force that is turned into electrical energy.

$$F_t = \begin{cases} F_{\text{drive}}, & \text{acceleration} \\ -\eta_{\text{regen}} F_{\text{brake}}, & \text{regenerative braking} \end{cases} \quad (8)$$

The complete force equation after substituting all derived components can be rewritten as:

$$\delta M \frac{dv}{dt} = F_t - \left[MgC_r \cos \theta + Mg \sin \theta + \frac{1}{2} \rho C_d A_f (v_e + v_{\text{wind}} \cos \phi)^2 \right] \quad (9)$$

Once the force equation is determined, we can find the total energy needed for an EV to travel the road segment by integrating force along the direction of motion:

$$E = \int F ds \quad (10)$$

Multiplying both sides of (1) by the differential displacement ds and integrating over a road segment of length l_e :

$$\int_0^{l_e} F_t ds = \int_0^{l_e} \delta M \frac{dv}{dt} ds + \int_0^{l_e} (F_{\text{roll}} + F_{\text{grade}} + F_{\text{drag}}) ds \quad (11)$$

The left side of the equation shows the energy generated from traction, and the right side includes changes in kinetic energy and energy losses from resistive forces. Integrating all parts, the total energy needed to traverse edge e is:

$$E_e = \frac{\frac{1}{2} \delta M (v_b^2 - v_c^2) + MgC_r l_e \cos \theta + Mgl_e \sin \theta}{\eta_{\text{drive}}} + \frac{\frac{1}{2} \rho C_d A_f v_{\text{rel}}^2 l_e}{\eta_{\text{drive}}} + E_{\text{aux}} - E_{\text{regen}} \quad (12)$$

The first term represents the energy associated with acceleration or deceleration, wherein v_b and v_c denote vehicle velocities at the beginning and end of the segment, respectively. The second term is rolling resistance energy, third term represents gravitational energy and fourth term is energy due to aerodynamic losses. Energy losses are divided by the drivetrain efficiency η_{drive} to account for conversion losses.

On steep downhill roads, the energy recovered from braking can exceed the energy losses, resulting in a negative net energy E_e . Auxiliary energy consumption for HVAC systems and onboard electronics is given by

$$E_{aux} = P_{aux} \cdot t_e \quad (13)$$

The energy during negative traction, which can be recovered as electrical energy, is given by:

$$E_{\text{regen}} = -\eta_{\text{regen}} \int_{F_t < 0} F_t ds \quad (14)$$

From (12), it can be observed that rolling resistance and drag are losses and cannot be recovered. Hence, the gravitational ($\theta < 0$) and kinetic energy ($\frac{dv}{dt} < 0$) will contribute to the regenerative energy.

$$E_{\text{regen}} = \eta_{\text{regen}} (|E_{\text{grade}}| + |E_{\text{kin}}|) \quad (15)$$

Once the energy model is established, we now compute the total time required to travel the road segment. Real-world traffic conditions impact both travel time and energy use due to changes in speed and the stop-and-go driving patterns. The actual traversal time must include traffic-induced speed reductions:

$$t_e = \frac{l_e}{s_e \cdot \tau_e} \quad (16)$$

where $\tau_e \in (0, 1]$ represents the speed reduction factor due to congestion. Third, the economic aspect of path selection introduces the cost constraint, given as:

$$C_e = E_e \cdot p_{elec} \quad (17)$$

where p_{elec} is the electricity price per kilowatt-hour. Because electricity costs are generally lower than those of conventional fuels, this factor becomes significant for minimizing operating expenses.

After defining the objective functions, it is important to introduce operational constraints to ensure route feasibility. The main constraint is the battery SOC. This represents the remaining energy in the battery, shown as a percentage of its total capacity, defined as:

$$SOC_{k+1} = SOC_k - \frac{E_e}{E_{bat}} \quad (18)$$

where E_{bat} is the total battery capacity, and SOC_k represents the state-of-charge at vertex k . To protect the battery from deep discharge and ensure sufficient reserve for unexpected delays, a minimum SOC must be maintained:

$$SOC_k \geq SOC_{min} \quad (19)$$

Any path violating this constraint at any junction is rejected during the optimal path search process.

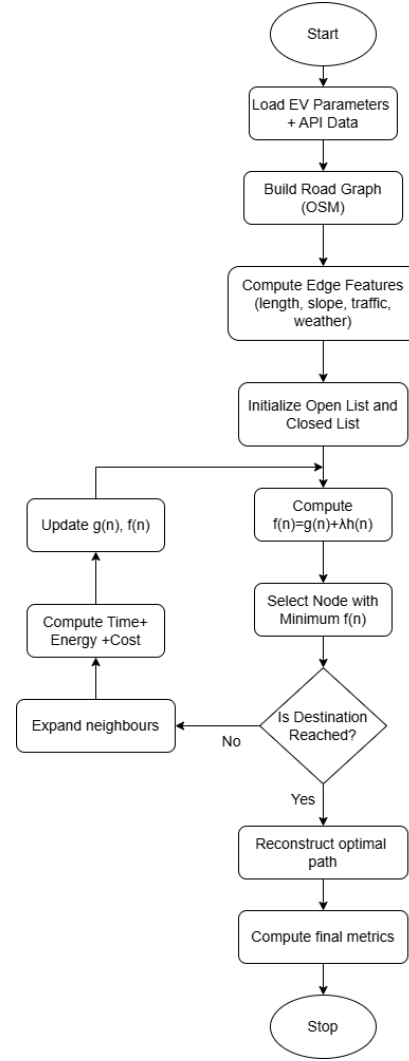


Fig. 2. Flow chart of proposed method

B. Optimisation Strategy

After developing the objective functions and specifying the operational constraints, we now formulate the optimisation strategy. Given a source node $s \in V$ and a destination node $t \in V$ in the road network graph G , the goal is to find a feasible route $P = \{e_1, e_2, \dots, e_n\}$ that optimises the objective functions while meeting all constraints. For this, we define an MO cost function that adds the costs of all the objectives across all road segments in path P :

$$J(P) = \sum_{e \in P} (w_1 \hat{E}_e + w_2 \hat{t}_e + w_3 \hat{C}_e + w_4 \hat{l}_e) \quad (20)$$

The variables $\hat{E}_e$, $\hat{t}_e$, $\hat{C}_e$, and $\hat{l}_e$ represent the normalised values of energy consumption, travel time, cost, and distance, respectively. The coefficients w_1 , w_2 , w_3 , and w_4 are weighting factors that determine the importance of each objective. As all quantities have different units and scales, we normalise them using the min-max method:

$$\hat{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (21)$$

This transformation changes all values to a common scale between 0 and 1, ensuring no objective dominates because it has larger numbers.

$$\sum_{i=1}^4 w_i = 1 \quad (22)$$

This constraint ensures that the total cost is properly scaled. The optimisation problem is now stated as finding a path P^* that minimizes the total cost:

Distance-optimal Dijkstra minimizes the overall distance $\min \sum l_e$ while ignoring energy costs and terrain effects. Steep hills on a shorter route may require more energy than a longer one. Time-optimal Dijkstra reduces travel time $\min \sum t_e$ but may lead to high-speed routes with air drag. k -Shortest Path Search (Yen's Algorithm) selects the best path P^* from a set of generated paths $\{P_1, \dots, P_k\}$, but it may not be the energy-optimal path. Bellman-Ford's $\mathcal{O}(|V||E|)$ complexity makes it unsuitable for quick navigation, even though it considers negative cost from RB. Therefore, we require a solution that can handle negative costs while maintaining efficiency.

Hence, the weighted A* algorithm is used as it addresses both requirements. It is an enhanced version of the standard A* pathfinding method, given as:

$$f(n) = g(n) + h(n) \quad (23)$$

where $g(n)$ is the total accumulated MO cost from the start to the current node n . This is calculated by adding all edge costs from (19), given as:

$$g(n) = \sum_{e \in P_{s \rightarrow n}} (w_1 \hat{E}_e + w_2 \hat{t}_e + w_3 \hat{C}_e + w_4 \hat{l}_e) \quad (24)$$

The term $h(n)$ is a heuristic function that estimates the cost from the current node to the destination. Mathematically, for any given node n , the heuristic estimate must satisfy the condition $h(n) \leq h^*(n)$, where $h^*(n)$ is the actual cost to the goal, making it admissible by never overestimating the true cost. For $h(n)$, we use the straight line (Euclidean) distance written as:

$$h(n) = \sqrt{(x_n - x_t)^2 + (y_n - y_t)^2} \quad (25)$$

where (x_n, y_n) and (x_t, y_t) are the coordinates of node n and destination t , respectively. This heuristic works well because the straight-line distance is always less than or equal to the actual distance of the road segment. The algorithm uses this value to select the next node based on the lowest $f(n)$ value, thus ensuring an optimal balance between known costs and estimated remaining distances.

$$P^* = \arg \min_P J(P) \quad (26)$$

subject to the battery state-of-charge constraint from Equation (18). To solve this problem, we compare various algorithms

Algorithm 1: Weighted A* MO EV Route Planning

Output: Optimal path P^* from source s to destination t
Initialize OPEN priority queue and CLOSED set;
Initialize $g(s) = 0$, $SOC(s) = SOC_0$;
Compute $f(s) = g(s) + \lambda h(s)$;
Insert s into OPEN;
while OPEN is not empty **do**
 Select node n with minimum $f(n)$ from OPEN;
 if $n == t$ **then**
 Reconstruct P^* using parent pointers;
 Return P^* ;
 Move n from OPEN to CLOSED;
 foreach neighbor v of n **do**
 Compute MO edge cost $c_{n,v}$;
 Compute energy consumption $E_{n,v}$;
 Update battery state $SOC_v = SOC_n - \frac{E_{n,v}}{E_{bat}}$;
 if $SOC_v < SOC_{min}$ **then**
 Skip node v ;
 Compute tentative cost $g_{cand} = g(n) + c_{n,v}$;
 if $g_{cand} < g(v)$ **then**
 Update $g(v) = g_{cand}$;
 Update $SOC(v) = SOC_v$;
 Set parent(v) = n ;
 Compute $f(v) = g(v) + \lambda h(v)$;
 Insert or update v in OPEN;
Return No feasible route found;

to identify their limitations and then select the most effective approach.

To achieve faster convergence, we employ the weighted A* algorithm with a weighting parameter λ :

$$f(n) = g(n) + \lambda h(n) \quad (27)$$

By setting $\lambda > 1$, more importance is given to the heuristic function to speed up the search significantly. When $\lambda = 1$, we have the standard A* algorithm, which guarantees finding the best path. When $\lambda > 1$ (here taken as 1.2), the algorithm finds a near-optimal path that might be a bit longer, but it does so much faster. We present a pseudocode of our energy-optimal weighted A* search with energy constraints in Algorithm 1.

TABLE II
PERFORMANCE COMPARISON OF ALGORITHMS

Algorithm	Energy (kWh)	Time (min)	Cost (USD)	Dist (km)	MO Cost	Runtime (ms)	SOC (%)
Distance-Opt	1.93	11.01	0.39	3.53	1.58	11.07	76.78
Time-Opt	1.76	10.12	0.35	3.66	1.45	10.09	77.09
Weighted A*	0.92	12.77	0.18	4.23	1.07	27.39	78.47
Yen's	1.05	11.68	0.21	3.82	1.10	371.18	78.25

In Fig. 3, the heat map shows variations in traffic density and the efficiency of the proposed algorithm in avoiding high-density zones. Traffic density heatmap and EV charging stations are rendered from live OSM and Overpass API data, respectively. Fig. 4 shows that weighted A* yields the lowest MO cost among the algorithms.

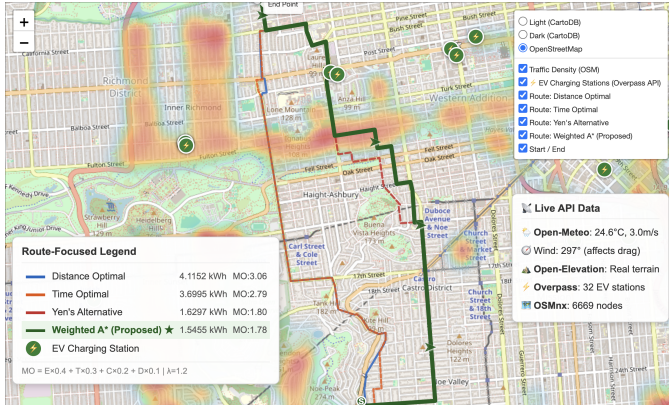


Fig. 3. Multi-objective EV route comparison across four algorithms on the San Francisco road network (OSMnx/OSM)

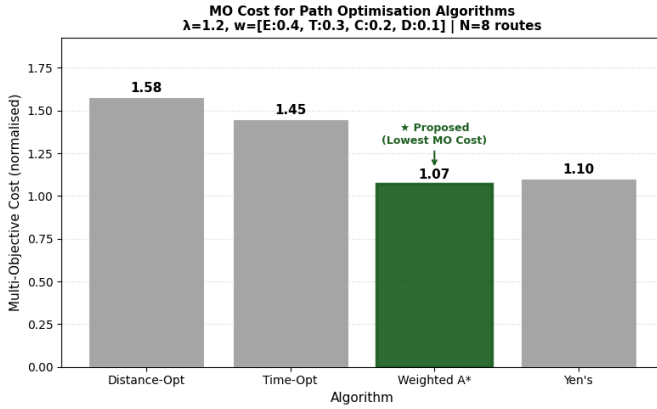


Fig. 4. MO Cost for Path Optimisation Algorithm

III. CONCLUSION

The paper presents an MO route optimisation methodology for EVs using the weighted A* algorithm. The road network is modelled as a directed weighted graph, where each edge cost is calculated by a physics-based model considering rolling resistance, aerodynamic drag, terrain changes, and RB. Energy consumption, travel time, cost and distance are normalised and combined into a weighted MO function, considering battery SOC constraints to ensure route feasibility. The proposed Weighted A* achieves MO cost reductions of 32.3%, 26.2%, and 2.7% over Distance-Optimal, Time-Optimal Dijkstra, and Yen's algorithm respectively (Table II), depicting its superiority across all baseline comparisons. In future work, we aim to integrate OCPP charging protocols, real-time rerouting and multi-vehicle scenarios for real-world applications.

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